

Aamir Hussain

Silicon Automation & Validation Engineer

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Professional Summary

Silicon Validation Engineer with 4 years of experience in pre-silicon and post-silicon validation, platform bring-up, and Python test automation for Intel Xeon server platforms and Microsoft Maia AI accelerators. Automated 300+ test cases using Python, Robot Framework, reducing validation cycle time by 60%. Hands-on with PCIe, DDR4/DDR5, SR-IOV, firmware validation, and hardware/firmware debug using JTAG, ITP and logic analyzers.

Technical Skills

- **Silicon Validation:** Post-Silicon Validation, Pre-Silicon Validation, Silicon Bring-Up, System-Level Validation, Functional Testing, Regression Testing, Failure Analysis, Root Cause Analysis
- **Interfaces & Protocols:** PCIe Gen4/Gen5/Gen6, SR-IOV, SerDes, High-Speed I/O, Ethernet, USB
- **Memory:** DDR4, DDR5, HBM, Memory Controller/PHY, Memory Training, Margin Testing, Stress Testing
- **Automation & Programming:** Python, Robot Framework, Test Automation, Regression Automation, Linux
- **Firmware & Debug:** Firmware Validation, JTAG, ITP, Logic Analyzers, Register-Level Debugging, Log and Trace Analysis, Firmware Flashing
- **Platforms:** Intel Xeon (Sapphire Rapids, Emerald Rapids, Granite Rapids), AI Accelerators (Microsoft Maia 200/300), Server Validation, Virtualization, Power and Thermal Validation

Professional Experience

Python Automation Engineer (Silicon Validation) | UST Global | Bengaluru, India | Oct 2022 - Present

Project: Microsoft AI Accelerator Card (Maia 200 / Maia 300), Client: Microsoft | 1 year

- Owned end-to-end validation and automation for Microsoft Maia 200 and Maia 300 AI accelerator platforms.
- Developed Python and Robot Framework automation frameworks for HSP firmware validation and regression testing.
- Implemented SR-IOV virtualization and configured PF/VF/VM hierarchies to enable parallel VM communication for AI inference workloads.
- Created secure authentication and communication scripts for HCP, ECP and PCP subsystems.
- Analyzed the BEN networking fabric to optimize low-latency, high-bandwidth AI inference performance.
- Authored technical documentation for system workflows and secure firmware authentication to support knowledge transfer.

Project: Intel Xeon Server Platform (Sapphire Rapids, Emerald Rapids, Granite Rapids), Client: Intel | 3 years

- Executed pre-silicon and post-silicon validation covering PMSS, PCIe, DDR, RAPL, P-States, C-States, SST, PECL and ISS.
- Designed Python-based automation and built regression pipelines, automating 300+ test cases and reducing validation cycle time by 60%.
- Validated 1S/2S/4S/8S multi-socket server configurations and optimized DIMM setups for performance.
- Debugged hardware and firmware issues using JTAG, ITP and logic analyzers to resolve complex system failures.
- Performed firmware flashing and platform bring-up, and documented bring-up procedures for the team.

Education

B.Tech, Computer Science and Engineering | Jammu University | 2021